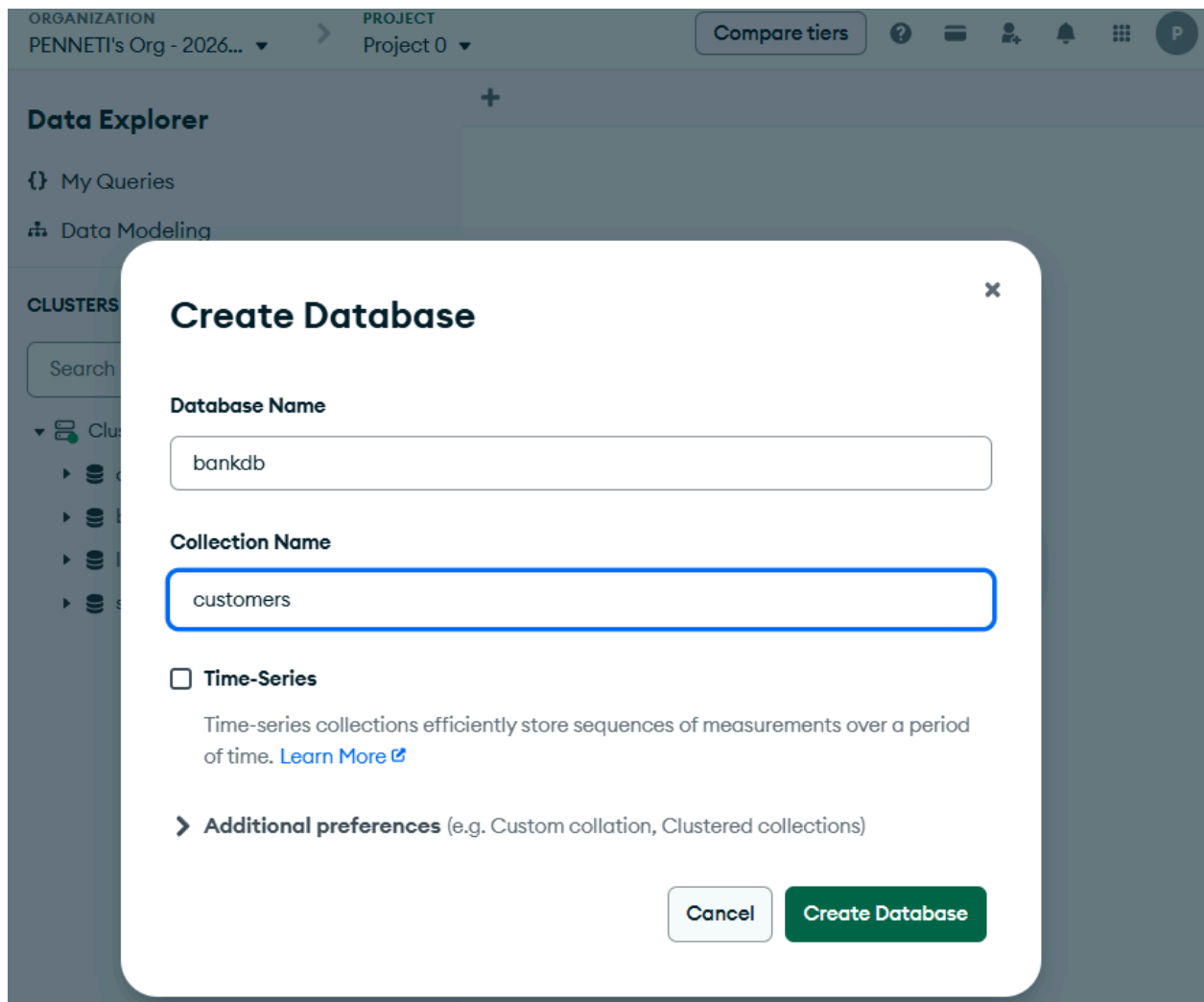


Database Skill Session 4 and 5

Name: P.Bindhu Madhavi
Roll Number: 2520030077
Section: 7

STEP 1: Create Database



The screenshot shows the MongoDB Cloud Portal interface. At the top, there's a header with 'ORGANIZATION' (PENNETI's Org - 2026...) and 'PROJECT' (Project 0). A 'Compare tiers' button is visible. Below the header, the 'Data Explorer' section is active, showing 'My Queries' and 'Data Modeling'. A 'CLUSTERS' sidebar is on the left. A modal dialog titled 'Create Database' is open in the center. It has a close button (X) in the top right corner. The dialog contains two text input fields: 'Database Name' with the value 'bankdb' and 'Collection Name' with the value 'customers'. Below these fields, there is a checkbox for 'Time-Series' which is currently unchecked. A description for Time-Series collections is provided: 'Time-series collections efficiently store sequences of measurements over a period of time. [Learn More](#)'. At the bottom of the dialog, there is a section for 'Additional preferences' (e.g. Custom collation, Clustered collections) with a right-pointing arrow. At the very bottom of the dialog are two buttons: 'Cancel' and 'Create Database'.

ORGANIZATION
PENNETI's Org - 2026... ▾

PROJECT
Project 0 ▾

Compare tiers

Data Explorer

My Queries

Data Modeling

CLUSTERS

Search

Cluster

Database Name
bankdb

Collection Name
customers

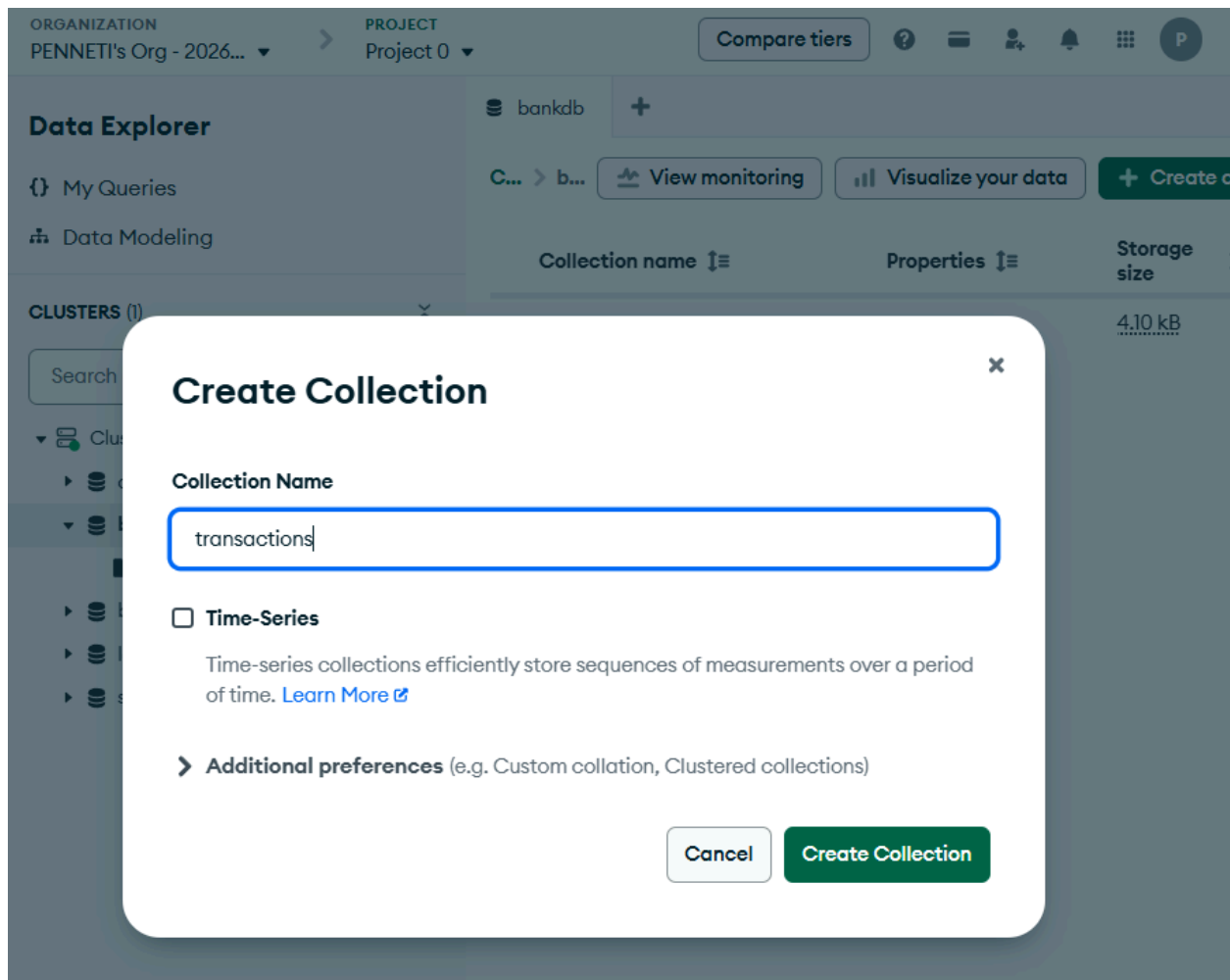
☐ Time-Series

Time-series collections efficiently store sequences of measurements over a period of time. [Learn More](#)

> Additional preferences (e.g. Custom collation, Clustered collections)

Cancel Create Database

STEP 2: Create Collection



STEP 3: Insert Customer 1 data

ORGANIZATION
PENNETI's Org - 2026...

PROJECT
Project 0

Compare tiers

Data Explorer

bankdb

customers



My Queries

Cluster0 > bankdb > customers

View monitoring

Data

CLUSTERS

Search

Clusters

Cluster0

Cluster1

Cluster2

Cluster3

Cluster4

Cluster5

Cluster6

Cluster7

Cluster8

Cluster9

Cluster10

Cluster11

Cluster12

Cluster13

Cluster14

Cluster15

Cluster16

Cluster17

Cluster18

Cluster19

Cluster20

Cluster21

Cluster22

Cluster23

Cluster24

Cluster25

Cluster26

Cluster27

Cluster28

Cluster29

Cluster30

Cluster31

Cluster32

Cluster33

Insert Document

To collection bankdb.customers

VIEW



```
1 {  
2   "customer_id": 101,  
3   "name": "Ravi Kumar",  
4   "city": "Hyderabad",  
5   "account_type": "Savings",  
6   "balance": 85000,  
7   "status": "Active"  
8 }  
9 |
```

Cancel

Insert

Data Explorer



- customers

+

{ } My Queries

Cluster0 > bankdb > customers

[View monitoring](#)

- Data M

Indexes	1	Val
---------	---	-----

CLUSTERS

Search

▼ Clu

Insert Document

To collection bankdb.customers

i VIEW

```
1 {
2   "customer_id": 102,
3   "name": "Priya Sharma",
4   "city": "Vijayawada",
5   "account_type": "Current",
6   "balance": 125000,
7   "status": "Active"
8 }
9
```



Cancel

Insert

ORGANIZATION
PENNETI's Org - 2026...

PROJECT
Project 0

Compare tiers

Data Explorer

bankdb

customers

My Queries

Cluster0 > bankdb > customers

View monitoring

Data M

Indexes 1

CLUSTERS

Search

Clu

Insert Document

To collection bankdb.customers

VIEW



```
1 {  
2   "customer_id": 103,  
3   "name": "Arjun Reddy",  
4   "city": "Hyderabad",  
5   "account_type": "Savings",  
6   "balance": 45000,  
7   "status": "Active"  
8 }  
9  
10
```

Cancel

Insert

ORGANIZATION
PENNETI's Org - 2026...

PROJECT
Project 0

Compare tiers

P

bankdb

customers

+

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

admin

bankdb

customers

transactions

bookflow

local

sample_mflix

Cluster0 > bankdb > customers

View monitoring

Documents 3

Aggregations

Schema

Indexes 1

Valid

Explain

Reset

25

1 - 3 of 3

balance : 85000

status : "Active"

_id: ObjectId('6a8296f0b84ecf52eabd429b')

customer_id : 102

name : "Priya Sharma"

city : "Vijayawada"

account_type : "Current"

balance : 125000

status : "Active"

_id: ObjectId('6a82975fb84ecf52eabd429d')

customer_id : 103

name : "Arjun Reddy"

city : "Hyderabad"

account_type : "Savings"

balance : 45000

status : "Active"

STEP 6: Insert Customer 4 data

Insert Document

To collection bankdb.customers

VIEW [JSON icon] [Menu icon]

```
1 {
2   "customer_id": 104,
3   "name": "Sneha Rao",
4   "city": "Warangal",
5   "account_type": "Savings",
6   "balance": 210000,
7   "status": "Active"
8 }
9
10
```

Cancel **Insert**

ORGANIZATION
PENNETI's Org - 2026...

PROJECT
Project 0

Compare tiers

P

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

admin

bankdb

customers

transactions

bookflow

local

sample_mflix

bankdb

customers

+

Cluster0 > bankdb > customers

View monitoring

Documents 4

Aggregations

Schema

Indexes 1

Vali

Explain

Reset

25

1 - 4 of 4

balance : 125000

status : "Active"

_id: ObjectId('6a82975fb84ecf52eabd429d')

customer_id : 103

name : "Arjun Reddy"

city : "Hyderabad"

account_type : "Savings"

balance : 45000

status : "Active"

_id: ObjectId('6a8297cfb84ecf52eabd429f')

customer_id : 104

name : "Sneha Rao"

city : "Warangal"

account_type : "Savings"

balance : 210000

status : "Active"

STEP 7: Insert Customer 5 data

The screenshot shows the MongoDB Data Explorer interface. A modal dialog titled "Insert Document" is open, showing the target collection as "bankdb.customers". The document to be inserted is a JSON object with the following fields:

```
1
2 {
3   "customer_id": 105,
4   "name": "Kiran Kumar",
5   "city": "Guntur",
6   "account_type": "Current",
7   "balance": 65000,
8   "status": "Inactive"
9 }
10
```

The dialog has a "Cancel" button and an "Insert" button at the bottom right.

ORGANIZATION
PENNETI's Org - 2026... ▾

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Project 0 ▾

Compare tiers ? P

Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Cluster0

admin

bankdb

customers

transactions

bookflow

local

sample_mflix

bankdb customers +

Cluster0 > bankdb > customers [View monitoring](#)

Documents 5

Aggregations

Schema

Indexes 1

Validations

1 - 5 of 5

[Explain](#) [Reset](#)

balance : 45000

status : "Active"

```
_id: ObjectId('6a8297cfb84ecf52eabd429f')
customer_id: 104
name: "Sneha Rao"
city: "Warangal"
account_type: "Savings"
balance: 210000
status: "Active"
```

```
_id: ObjectId('6a82985db84ecf52eabd42a1')
customer_id: 105
name: "Kiran Kumar"
city: "Guntur"
account_type: "Current"
balance: 65000
status: "Inactive"
```

STEP 8: Insert Customer 6 data

ORGANIZATION
PENNETI's Org - 2026...

PROJECT
Project 0

Compare tiers

Data Explorer

bankdb

customers

My Queries

Cluster0 > bankdb > customers

View monitoring

Data Explorer

CLUSTERS

Search

Insert Document

To collection bankdb.customers

VIEW

JSON

```
1 {  
2   "customer_id": 106,  
3   "name": "Anjali Singh",  
4   "city": "Hyderabad",  
5   "account_type": "Savings",  
6   "balance": 175000,  
7   "status": "Active"  
8 }  
9  
10
```

Cancel

Insert

ORGANIZATION
PENNETI's Org - 2026... ▾

PROJECT
Project 0 ▾

Compare tiers

?

P

Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bankdb
 - customers
 - transactions
- bookflow
- local
- sample_mflix

bankdbcustomers+

Cluster0 > bankdb > customers

View monitoring

Documents 6AggregationsSchemaIndexes 1Validations

🕒 ▾

▶ ▾

Explain

Reset

+ ▾

👤 ▾

</>

25 ▾

1 - 6 of 6

↺

↻

↷

balance : 210000

status : "Active"

_id: ObjectId('6a82985db84ecf52eabd42a1')

customer_id : 105

name : "Kiran Kumar"

city : "Guntur"

account_type : "Current"

balance : 65000

status : "Inactive"

_id: ObjectId('6a8298c2b84ecf52eabd42a3')

customer_id : 106

name : "Anjali Singh"

city : "Hyderabad"

account_type : "Savings"

balance : 175000

status : "Active"

STEP 9: Insert Customer 7 data

ORGANIZATION
PENNETI's Org - 2026...

PROJECT
Project 0

Compare tiers

P

Data Explorer

bankdbcustomers+

Cluster0 > bankdb > customersView monitoring

My QueriesData MCLUSTERS

Search

Clu

Insert Document

X

To collection bankdb.customers

iVIEW

{}

1{

2"customer_id": 107,

3"name": "Rahul Verma",

4"city": "Bengaluru",

5"account_type": "Current",

6"balance": 95000,

7"status": "Active"

8}

Cancel

Insert

ORGANIZATION
PENNETI's Org - 2026...

PROJECT
Project 0

Compare tiers

P

Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

admin

bankdb

customers

transactions

bookflow

local

sample_mflix

bankdb

customers

Cluster0 > bankdb > customers

View monitoring

Documents 7AggregationsSchemaIndexes 1Va

Explain

Res

25

1 - 7 of 7

balance : 65000

status : "Inactive"

_id: ObjectId('6a8298c2b84ecf52eabd42a3')

customer_id : 106

name : "Anjali Singh"

city : "Hyderabad"

account_type : "Savings"

balance : 175000

status : "Active"

_id: ObjectId('6a829937b84ecf52eabd42a5')

customer_id : 107

name : "Rahul Verma"

city : "Bengaluru"

account_type : "Current"

balance : 95000

status : "Active"

STEP 10: Insert Customer 8 data

Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

admin

bankdb

customers

transactions

bookflow

local

sample_mflix

bankdb customers transactions +

Cluster0 > bankdb > transactions

Documents 15 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA BULK EXPORT CODE

```
_id: ObjectId('6a829a32b84ecf52eabd42aa')
transaction_id: 1001
customer_id: 101
type: "Deposit"
amount: 25000
mode: "UPI"
date: "2026-08-01"
status: "Success"
```

```
_id: ObjectId('6a829a41b84ecf52eabd42ac')
transaction_id: 1002
customer_id: 101
type: "Withdrawal"
amount: 10000
mode: "ATM"
date: "2026-08-02"
status: "Success"
```

Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

admin

bankdb

customers

transactions

bookflow

local

sample_mflix

bankdb customers transactions +

Cluster0 > bankdb > transactions

Documents 15 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA BULK EXPORT CODE

```
_id: ObjectId('6a829a4fb84ecf52eabd42ae')
transaction_id: 1003
customer_id: 102
type: "Deposit"
amount: 50000
mode: "NEFT"
date: "2026-08-02"
status: "Success"
```

```
_id: ObjectId('6a829a60b84ecf52eabd42b0')
transaction_id: 1004
customer_id: 102
type: "Withdrawal"
amount: 15000
mode: "ATM"
date: "2026-08-03"
status: "Success"
```

ORGANIZATION
PENNETI's Org - 2026...

PROJECT
Project 0

Data Explorer

{ } My Queries

🔗 Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

admin

bankdb

customers

transactions

bookflow

local

sample_mflix

bankdb

customers

transactions



Cluster0 > bankdb > transactions

Documents 15

Aggregations

Schema

Indexes 1

Validation



Type a query: { field: 'value' } or [Generate query](#)

ADD DATA

BULK

EXPORT CODE

```
_id: ObjectId('6a829a6cb84ecf52eabd42b2')
transaction_id: 1005
customer_id: 103
type: "Deposit"
amount: 30000
mode: "UPI"
date: "2026-08-03"
status: "Success"
```

```
_id: ObjectId('6a829a79b84ecf52eabd42b4')
transaction_id: 1006
customer_id: 103
type: "Withdrawal"
amount: 5000
mode: "ATM"
date: "2026-08-04"
status: "Success"
```

Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

admin

bankdb

customers

transactions

bookflow

local

sample_mflix

bankdb

customers

transactions



Cluster0 > bankdb > transactions

Documents 15

Aggregations

Schema

Indexes 1

Validation



Type a query: { field: 'value' } or [Generate query](#)

ADD DATA

BULK

EXPORT CODE

```
_id: ObjectId('6a829a85b84ecf52eabd42b6')
transaction_id: 1007
customer_id: 104
type: "Deposit"
amount: 100000
mode: "NEFT"
date: "2026-08-04"
status: "Success"
```

```
_id: ObjectId('6a829a91b84ecf52eabd42b8')
transaction_id: 1008
customer_id: 104
type: "Withdrawal"
amount: 25000
mode: "ATM"
date: "2026-08-05"
status: "Success"
```

Data Explorer

{ } My Queries

🔗 Data Modeling

CLUSTERS (1)

Cluster0

admin

bankdb

customers

transactions

bookflow

local

sample_mflix

bankdb

customers

transactions



Cluster0 > bankdb > transactions

Documents 15

Aggregations

Schema

Indexes 1

Validation

Type a query: { field: 'value' } or [Generate query](#)

ADD DATA

BULK

EXPORT CODE

```
_id: ObjectId('6a829aaab84ecf52eabd42bc')
transaction_id: 1010
customer_id: 106
type: "Deposit"
amount: 75000
mode: "NEFT"
date: "2026-08-06"
status: "Success"
```

```
_id: ObjectId('6a829ab6b84ecf52eabd42be')
transaction_id: 1011
customer_id: 106
type: "Withdrawal"
amount: 20000
mode: "ATM"
date: "2026-08-06"
status: "Success"
```

ORGANIZATION
PENNETI's Org - 2026... ▾

PROJECT
Project 0 ▾

Data Explorer

{ } My Queries

👤 Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

admin

bankdb

customers

transactions

bookflow

local

sample_mflix

bankdb

customers

transactions



Cluster0 > bankdb > transactions

Documents 15

Aggregations

Schema

Indexes 1

Validation



Type a query: { field: 'value' } or [Generate query](#) ✨

+ ADD DATA ▾

BULK ▾

EXPORT CODE

```
_id: ObjectId('6a829ac1b84ecf52eabd42c0')
transaction_id: 1012
customer_id: 107
type: "Deposit"
amount: 40000
mode: "IMPS"
date: "2026-08-07"
status: "Success"
```

```
_id: ObjectId('6a829aceb84ecf52eabd42c2')
transaction_id: 1013
customer_id: 107
type: "Withdrawal"
amount: 12000
mode: "ATM"
date: "2026-08-07"
status: "Success"
```

Data Explorer

{ } My Queries

🔗 Data Modeling

CLUSTERS (1)

- Cluster0
 - admin
 - bankdb
 - customers
 - transactions**
 - bookflow
 - local
 - sample_mflix

bankdb customers transactions +

Cluster0 > bankdb > transactions

Documents 15 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

+ ADD DATA

BULK

EXPORT CODE

status : "Success"

```
_id: ObjectId('6a829b69b84ecf52eabd42c4')
transaction_id: 1014
customer_id: 108
type: "Deposit"
amount: 150000
mode: "NEFT"
date: "2026-08-08"
status: "Success"
```

```
_id: ObjectId('6a829b76b84ecf52eabd42c6')
transaction_id: 1015
customer_id: 108
type: "Withdrawal"
amount: 30000
mode: "ATM"
date: "2026-08-08"
status: "Success"
```

STEP 12: Find Active Customer

ORGANIZATION PENNETI's Org - 2026... PROJECT Project 0

Compare tiers

Data Explorer

My Queries
Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - admin
 - bankdb
 - customers
 - transactions
 - bookflow
 - local
 - sample_mflix

bankdb customers transactions

Cluster0 > bankdb > customers View monitoring

Documents 8 Aggregations Schema Indexes 1 Validate

```
{  "status": "Active"}
```

Explain Reset

25 1 - 7 of 7

```
_id: ObjectId('6a829937b84ecf52eabd42a5')
customer_id: 107
name: "Rahul Verma"
city: "Bengaluru"
account_type: "Current"
balance: 95000
status: "Active"
```

```
_id: ObjectId('6a829989b84ecf52eabd42a7')
customer_id: 108
name: "Meena Devi"
city: "Chennai"
account_type: "Savings"
balance: 320000
```

STEP 13: Find Customers with Balance Greater Than 100000

ORGANIZATION
PENNETI's Org - 2026...>

PROJECT
Project 0

Compare tiers?

P

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

admin

bankdb

customers

transactions

bookflow

local

sample_mflix

Cluster0 > bankdb > customers

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Documents 8

Aggregations

Schema

Indexes 1

Validations

{

"balance": {

"\$gt": 100000

}

}

Explain

Reset

25

1 - 4 of 4

_id: ObjectId('6a8298c2b84ecf52eabd42a3')

customer_id: 106

name: "Anjali Singh"

city: "Hyderabad"

account_type: "Savings"

balance: 175000

status: "Active"

_id: ObjectId('6a829989b84ecf52eabd42a7')

customer_id: 108

name: "Meena Devi"

city: "Chennai"

account_type: "Savings"

balance: 320000

STEP 14: Sort Customers By Balance(\$sort)

ORGANIZATION PROJECT

Stage 1: \$sort > ENABLED < ADD STAGE

STAGE INPUT

\$sort [Open docs](#)

```
1 ▾ /**
2  * Provide any number of field/order pairs.
3  */
4 ▾ {
5    balance: -1
6  }
```

No preview documents

STAGE OUTPUT

Sample of 8 documents [OPTIONS](#)

```
_id: ObjectId('6a82989b84ecf52eabd4...')
customer_id: 108
name: "Meena Devi"
city: "Chennai"
account_type: "Savings"
balance: 320000
status: "Active"

_id: ObjectId('6a8297cfb84ecf52eabd4...')
customer_id: 104
name: "Sneha Rao"
city: "Warangal"
account_type: "Savings"
balance: 210000
status: "Active"

_id: ObjectId('6a8298c2b84ecf52eabd4...')
customer_id: 106
name: "Anjali Singh"
city: "Hyderabad"
account_type: "Savings"
balance: 175000
status: "Active"

_id: ObjectId('6a8296f0b84ecf52eabd4...')
customer_id: 102
name: "Priya Sharma"
city: "Vijayawada"
account_type: "Current"
balance: 125000
status: "Active"
```

ORGANIZATION PROJECT

Stage 1: \$sort > ENABLED < ADD STAGE

STAGE INPUT

\$sort [Open docs](#)

```
1 ▾ /**
2  * Provide any number of field/order pairs.
3  */
4 ▾ {
5    balance: -1
6  }
```

No preview documents

STAGE OUTPUT

Sample of 8 documents [OPTIONS](#)

```
_id: ObjectId('6a829937b84ecf52eabd4...')
customer_id: 107
name: "Rahul Verma"
city: "Bengaluru"
account_type: "Current"
balance: 95000
status: "Active"

_id: ObjectId('6a82967bb84ecf52eabd4...')
customer_id: 101
name: "Ravi Kumar"
city: "Hyderabad"
account_type: "Savings"
balance: 85000
status: "Active"

_id: ObjectId('6a82985db84ecf52eabd4...')
customer_id: 105
name: "Kiran Kumar"
city: "Guntur"
account_type: "Current"
balance: 65000
status: "Inactive"

_id: ObjectId('6a82975fb84ecf52eabd4...')
customer_id: 103
name: "Arjun Reddy"
city: "Hyderabad"
account_type: "Savings"
balance: 45000
status: "Active"
```

STEP 15: CALCULATE TOTAL TRANSACTION AMOUNT(\$group)

ORGANIZATION
PENNETI's Org - 2026... > PROJECT
Project 0

Compare tiers ? [Menu] [User] [Bell] [More]

bankdb customers transactions +

Cluster0 > bankdb > customers View monitoring Visualize Your Data

Documents 8 Aggregations Schema Indexes 1 Validation

\$group Explain Run Options

Untitled - modified [Save] + CREATE NEW [Code Icon] PREVIEW [JSON Icon] [Code Icon] [Star] [Share]

Stage 1 \$group [Toggle]

```
1 {
2   "_id": null,
3   "total_amount": {
4     "$sum": "$amount"
5   }
6 }
7
8
```

Output preview after \$group stage (Sample of 1 document)

```
_id: null
total_amount : 0
```

STEP 16: Calculate Average Transaction Amount (\$group)

ORGANIZATION
PENNETI's Org - 2026...

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Project 0

Compare tiers

?

bankdbcustomerstransactions+

Cluster0 > bankdb > customers

View monitoring

Visualize Your Data

Documents 8AggregationsSchemaIndexes 1Validation

\$group

Explain

Run

Options

Untitled - modified

+ CREATE NEW

</>

PREVIEW

{ }

</>

+

Stage 1 \$group

1 {

2 "_id": null,

3 "average_amount": {

4 "\$avg": "\$amount"

5 }

6 }

7 |

8

Output preview after \$group stage (Sample of 1 document)

_id: null

average_amount : null

STEP 17: \$lookup

< Stage 1: \$lookup >

ENABLED

ADD STAGE

STAGE INPUT

\$lookup Open docs

```
1 * /**
2  * from: The target collection.
3  * localField: The local join field.
4  * foreignField: The target join field.
5  * as: The name for the results.
6  * pipeline: Optional pipeline to run on the foreign collection.
7  * let: Optional variables to use in the pipeline field stages.
8  */
9 * {
10  "from": "customers",
11  "localField": "customer_id",
12  "foreignField": "customer_id",
13  "as": "customer_details"
14 }
15
```

No preview documents

STAGE OUTPUT

OPTIONS

Sample of 10 documents

_id: ObjectId('6a829a32b84ecf52eabd4...')

transaction_id: 1001

customer_id: 101

type: "Deposit"

amount: 25000

mode: "UPI"

date: "2026-08-01"

status: "Success"

customer_details: Array (1)

_id: ObjectId('6a829a41b84ecf52eabd4...')

transaction_id: 1002

customer_id: 101

type: "Withdrawal"

amount: 10000

mode: "ATM"

date: "2026-08-02"

status: "Success"

customer_details: Array (1)

_id: ObjectId('6a829a4fb84ecf52eabd4...')

transaction_id: 1003

customer_id: 102

type: "Deposit"

amount: 50000

mode: "NEFT"

date: "2026-08-02"

status: "Success"

customer_details: Array (1)

< Stage 1: \$lookup >

ENABLED

ADD STAGE

STAGE INPUT

\$lookup Open docs

```
1 * /**
2  * from: The target collection.
3  * localField: The local join field.
4  * foreignField: The target join field.
5  * as: The name for the results.
6  * pipeline: Optional pipeline to run on the foreign collection.
7  * let: Optional variables to use in the pipeline field stages.
8  */
9 * {
10  "from": "customers",
11  "localField": "customer_id",
12  "foreignField": "customer_id",
13  "as": "customer_details"
14 }
15
```

No preview documents

STAGE OUTPUT

OPTIONS

Sample of 10 documents

_id: ObjectId('6a829a4fb84ecf52eabd4...')

transaction_id: 1003

customer_id: 102

type: "Deposit"

amount: 50000

mode: "NEFT"

date: "2026-08-02"

status: "Success"

customer_details: Array (1)

_id: ObjectId('6a829a60b84ecf52eabd4...')

transaction_id: 1004

customer_id: 102

type: "Withdrawal"

amount: 15000

mode: "ATM"

date: "2026-08-03"

status: "Success"

customer_details: Array (1)

_id: ObjectId('6a829a6cb84ecf52eabd4...')

transaction_id: 1005

customer_id: 103

type: "Deposit"

amount: 30000

mode: "UPI"

date: "2026-08-03"

status: "Success"

customer_details: Array (1)

Stage 1: \$lookup > ENABLED ADD STAGE ?

IE INPUT

\$lookup Open docs

```
1 /**
2  * from: The target collection.
3  * localField: The local join field.
4  * foreignField: The target join field.
5  * as: The name for the results.
6  * pipeline: Optional pipeline to run on the foreign collection.
7  * let: Optional variables to use in the pipeline field stages.
8  */
9 {
10   "from": "customers",
11   "localField": "customer_id",
12   "foreignField": "customer_id",
13   "as": "customer_details"
14 }
15
```

No preview documents

STAGE OUTPUT

Sample of 10 documents

OPTIONS

```
{
  "_id": ObjectId("6a829a79b84ecf52eabd4"),
  "transaction_id": 1006,
  "customer_id": 103,
  "type": "Withdrawal",
  "amount": 5000,
  "mode": "ATM",
  "date": "2026-08-04",
  "status": "Success",
  "customer_details": Array (1)
}
```

```
{
  "_id": ObjectId("6a829a85b84ecf52eabd4"),
  "transaction_id": 1007,
  "customer_id": 104,
  "type": "Deposit",
  "amount": 100000,
  "mode": "NEFT",
  "date": "2026-08-04",
  "status": "Success",
  "customer_details": Array (1)
}
```

```
{
  "_id": ObjectId("6a829a91b84ecf52eabd4"),
  "transaction_id": 1008,
  "customer_id": 104,
  "type": "Withdrawal",
  "amount": 25000,
  "mode": "ATM",
  "date": "2026-08-05",
  "status": "Success",
  "customer_details": Array (1)
}
```

Stage 1: \$lookup > ENABLED ADD STAGE ?

GE INPUT

\$lookup Open docs

```
1 /**
2  * from: The target collection.
3  * localField: The local join field.
4  * foreignField: The target join field.
5  * as: The name for the results.
6  * pipeline: Optional pipeline to run on the foreign collection.
7  * let: Optional variables to use in the pipeline field stages.
8  */
9 {
10   "from": "customers",
11   "localField": "customer_id",
12   "foreignField": "customer_id",
13   "as": "customer_details"
14 }
15
```

No preview documents

STAGE OUTPUT

Sample of 10 documents

OPTIONS

customer_details: Array (1)

```
{
  "_id": ObjectId("6a829a91b84ecf52eabd4"),
  "transaction_id": 1008,
  "customer_id": 104,
  "type": "Withdrawal",
  "amount": 25000,
  "mode": "ATM",
  "date": "2026-08-05",
  "status": "Success",
  "customer_details": Array (1)
}
```

```
{
  "_id": ObjectId("6a829a9eb84ecf52eabd4"),
  "transaction_id": 1009,
  "customer_id": 105,
  "type": "Deposit",
  "amount": 150000,
  "mode": "UPI",
  "date": "2026-08-05",
  "status": "Success",
  "customer_details": Array (1)
}
```

```
{
  "_id": ObjectId("6a829aaab84ecf52eabd4"),
  "transaction_id": 1010,
  "customer_id": 106,
  "type": "Deposit",
  "amount": 75000,
  "mode": "NEFT",
  "date": "2026-08-06",
  "status": "Success",
  "customer_details": Array (1)
}
```